Internship Report

On

Business Analytics

At

HANWOOL PVT. LTD.

Address: Plot No. – 284, Udyog Kendra, Ecotech-3, Ext-1, Greater Noida, G.B. Nagar, U.P. - 201306

BENNETT UNIVERSITY
TIMES OF INDIA GROUP

Session: 2022-26

Supervised by:

Mr. Arpit Gupta

(**Industry Mentor**)

and

Dr. Prabhishek Singh

(Batch Mentor)

Submitted by:

Shinde Tanmay Kailas

Bachelor of Technology (Computer Science and Engineering)

Enrolment No: E22CSEU0335

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND TECHNOLOGY,

BENNETT UNIVERSITY, GREATER NOIDA

CERTIFICATE

This is to certify that the student named Shinde Tanmay Kailas (Enrolment No: E22CSEU0335) of Programme Name Bachelor of Technology (Computer Science and Engineering), has undertaken an internship titled “Data Science” at HANWOOL PVT. LTD. From 10^{th} September 2025 to 10^{th} February 2026. This progress report is a bona fide record of the work completed during the period (10^{th} September 2025 to 26 September 2025).

Date: September 29, 2025

Mr. Prabhishek Singh

Internal Guide / Faculty Mentor (Name & Signature)

Mr. Arpit Gupta

Arpit

Industry Mentor (Name & Signature)

Internship Offer Letter & Joining Proof

GSTIN: 09AAGCH6535E1ZJ

HANWOOL PVT. LTD.

Add.: Plot No.- 284, Udyog Kendra, Ecotech-3, Ext - 1,
Greater Noida, G.B. Nagar, U.P. - 201306

Ref.:........................ *Date:*...........................

Shinde Tanmay Kailas
Address: Datta Nagar, Jamkhed, Ahilya Nagar, Maharashtra

Email: tanmayshinde32@gmail.com

Mob- +91 9503863289

Internship Offer – Data Science Intern at HANWOOL PVT LTD.

Dear Shinde Tanmay Kailas,

On behalf of the team at HANWOOL PVT LTD, I am delighted to extend to you an offer for the position of **Data Science Intern**. We were very impressed during the interview process and believe your skills and enthusiasm will be a valuable asset to our team.

This letter outlines the terms and conditions of your internship offer.

Position: Your official title will be **Data Science Intern**. You will be reporting directly to Arpit Gupta.

Start and End Dates: Your internship is scheduled to begin on **8th September 2025** and will conclude on **8th Feb 2026**

Work Schedule: This is a full-time internship. Your expected workdays will be Monday - Friday from **10:00 am to 06:00 pm** with a one-hour break for lunch.

Location: You will be working at our office located at Gautam buddha nagar, Uttar Pradesh.

Stipend / Compensation: **22,000 per month**

- Based on your performance and contributions during your tenure, you may be considered for a Pre-placement offer(PPO) with HANWOOL PVT LTD.

Internship Responsibilities: As a Data Science Intern, you will have the opportunity to work on meaningful projects and gain hands-on experience. Your primary responsibilities will include:

- Assist in the **collection, cleaning, and preprocessing** of datasets from multiple sources. Perform **exploratory data analysis (EDA)** to identify trends, patterns, and insights.
- Build and validate **statistical models or machine learning algorithms** under guidance. Create **visualizations, dashboards, and reports** to effectively communicate findings.

Company Policies: As an intern at HANWOOL PVT LTD, you will be expected to adhere to all company policies and procedures, including our code of conduct and confidentiality agreement, which you will be asked to sign on your first day.

Should you have any questions, please do not hesitate to contact me directly by phone at 9540497325.

We look forward to welcoming you to HANWOOL PVT LTD.

M/s HANWOOL PVT. LTD.
Sachin
Authorized Signatory

E-mail: sunil.kumar_22@rediffmail.com, jyotikumar427@gmail.com
Mob.: 9711708818, 9990866522, 9999280158, 7862002400

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout my internship at HANWOOL PVT. LTD.

I am deeply thankful to my industry mentor, Mr. Arpit Gupta for his invaluable guidance, continuous encouragement, and for providing me with the opportunity to work on meaningful projects. His insights have been instrumental in my learning and professional development.

I extend my gratitude to the entire team at HANWOOL PVT. LTD. for creating a welcoming and collaborative environment.

I am also grateful to my faculty mentor, Dr. Prabhishek Singh and the **School of Computer Science Engineering and Technology** at **Bennett University** for their academic support and for facilitating this internship as a part of our curriculum.

Finally, I would like to thank my family and friends for their unwavering support and encouragement throughout this journey.

Shinde Tanmay Kailas

E22CSEU0335

Chapter 1: Introduction of the Report

1.1 Background
This report documents the work and learning undertaken as part of the Data Science Internship at HANWOOL PVT. LTD. As a Fourth-year student pursuing a Bachelor of Technology in Computer Science and Engineering at Bennett University, this internship is a core component of my academic curriculum.

My academic background includes strong foundations in Data Structures & Algorithms and DBMS, complemented by practical expertise in Python (Pandas, NumPy, Scikit-learn), SQL, and data visualization tools. This internship provided me with the opportunity to bridge academic knowledge with real-world applications, gaining valuable exposure to data science workflows in a professional setting.

1.2 Purpose
The primary purpose of this internship is to gain hands-on experience in the end-to-end data science lifecycle and contribute to live projects at HANWOOL PVT. LTD. The key objectives include:

- To understand the practical application of data collection, cleaning, and preprocessing across diverse data sources.
- To perform exploratory data analysis (EDA) for identifying patterns, trends, and anomalies.
- To build and validate statistical models or machine learning algorithms under professional guidance.
- To enhance data storytelling skills through visualizations, dashboards, and reports.

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- To build and validate statistical models or machine learning algorithms under professional guidance.
- To enhance data storytelling skills through visualizations, dashboards, and reports.
- To contribute to the company's data-driven decision-making processes.

1.3 Technology Used / Data Collection Methods

During the internship, I am working with industry-standard tools and technologies, including:

- **Languages:** Python, SQL
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **Databases:** MySQL
- **Tools:** Jupyter Notebook, Microsoft Excel, Power BI
- **Version Control:** Git, GitHub

Data collection methods include direct SQL queries on company databases, structured datasets provided by the analytics team, and insights gathered from project documentation and team discussions.

1.4 Scope of the Internship

The scope of my role as a Data Science Intern encompasses multiple stages of the data science pipeline. My responsibilities include:

- Assisting in the collection, cleaning, and preprocessing of datasets from multiple sources.
- Conducting exploratory data analysis (EDA) to extract initial insights.
- Building and validating statistical models or machine learning algorithms under guidance.
- Creating dashboards, visualizations, and reports to effectively communicate findings.
- Adhering to company policies, confidentiality agreements, and professional standards.

Chapter 2: Overview of the Organization

2.1 Brief Overview of the Organization

HANWOOL PVT. LTD. is a dynamic and innovative technology company located in Greater Noida. The organization specializes in building robust software solutions and leveraging data-driven approaches to optimize business outcomes. With a forward-thinking outlook, HANWOOL emphasizes the role of analytics and artificial intelligence in modern decision-making, providing a fertile learning ground for aspiring data professionals.

2.2 Major Activities, Products/Services

The company's core activities revolve around software development, data analytics, and technology-driven business solutions. My work as a Data Science Intern contributes directly to the company's mission of harnessing data for actionable insights, whether in client services or internal efficiency improvements.

2.3 Hierarchical Structure
The Data Science team at HANWOOL PVT. LTD. works collaboratively, with a flat reporting structure that facilitates knowledge sharing and mentorship. As an intern, I report directly to my mentor, Mr. Arpit Gupta, a senior technical professional in the team. This arrangement ensures effective communication, timely feedback, and meaningful project contributions.

Chapter 3: What I Have Learned

3.1 Duration of Internship and Department Assigned

- **Duration:** The internship spans five months, beginning on 10th September 2025 and concluding on 10th February 2026.
- **Department:** I have been assigned to the Data Science team, under the supervision of my mentor, Mr. Arpit Gupta.

3.2 Responsibilities and Activities Performed
In the initial three weeks, my focus was on onboarding, understanding the existing data infrastructure, and contributing to data analysis tasks. The major activities included:

- **Environment Setup:** Installed and configured Python, Jupyter Notebook, Scikit-learn, and database connections.
- **Data Understanding:** Studied dataset schemas, business logic, and defined KPIs.
- **Data Collection & Cleaning:** Retrieved datasets using SQL queries and prepared them using Pandas, addressing missing values, duplicates, and inconsistencies.
- **Exploratory Data Analysis (EDA):** Applied descriptive statistics and visualizations (Matplotlib, Seaborn) to identify trends, correlations, and anomalies.
- **Initial Modeling:** Experimented with baseline machine learning models for predictive insights, gaining exposure to model validation techniques.

3.3 Skills Gained
This period has significantly enhanced both my technical and professional competencies:

- **Technical Skills:**
    - Learned to write advanced SQL queries for data extraction.
    - Improved proficiency in Python libraries for preprocessing and EDA.

    - Gained initial experience with machine learning workflows, including train-test splits and evaluation metrics.
    - Enhanced ability to create dashboards and visualizations for data storytelling.
- **Professional Skills:**
    - **Problem-Solving:** Developed skills to translate business requirements into data science tasks.
    - **Collaboration:** Actively participated in team discussions and learned effective professional communication.
    - **Time Management:** Balanced multiple tasks and delivered within deadlines.

3.4 Conclusion: Overall Experience and Personal Takeaways

The first phase of my internship at HANWOOL PVT. LTD. has been both insightful and rewarding. Working on real-world data problems with guidance from my mentor has allowed me to apply my academic knowledge practically while developing new technical and professional skills. This experience has boosted my confidence in working with datasets, performing EDA, and building initial models. I am enthusiastic about taking on more advanced machine learning tasks and contributing further in the coming months.

Chapter 4: Progress Summary

This summary covers the work accomplished during the first reporting period, from September 10, 2025, to September 27, 2025.

4.1 General Description of Tasks Done

My primary focus was integration into the team, understanding the data ecosystem, and contributing to an initial project. I performed data extraction using SQL, preprocessing in Python, exploratory data analysis, and initial model prototyping. I also participated in daily stand-ups and project planning discussions.

4.2 Milestones Achieved

- Successfully set up the complete working environment (Python, SQL, Jupyter).
- Collected and cleaned over 10,000 records for analysis using Pandas.
- Completed an exploratory data analysis with visualizations highlighting key patterns.
- Implemented baseline models (linear regression, decision trees) to test predictive performance.

- Delivered my first findings in a presentation to my mentor and received constructive feedback.

4.3 Skills/Tools Learned

- **Advanced SQL:** Queries with joins, aggregations, and subqueries.
- **Pandas & NumPy:** Efficient data cleaning and transformation.
- **Visualization:** Used Matplotlib and Seaborn for clear, data-driven storytelling.
- **Scikit-learn:** Gained hands-on experience in model building, training, and evaluation.
- **Jupyter Notebooks:** Documented complete workflows for reproducibility.

4.4 Reflections on Challenges Faced and Contributions Made

The biggest challenge was understanding the complex structure of raw data and aligning it with business requirements. I overcame this by building a personal data dictionary and seeking clarifications from my mentor. My contributions during this period—particularly a cleaned dataset and EDA-driven insights—helped accelerate the team's analysis. Furthermore, my baseline modeling results provided a foundation for more advanced models in the upcoming project phases.

Weekly Work Logs

Reporting Period: September 10, 2025 – September 27, 2025

Week No.	Dates (From–To)	Task Description	Hours Spent	Mentor Signature
12	Sep 15 – Sep 19	• Company and team onboarding. • Set up local analytics environment (Python, SQL client). • Initial review of business objectives and available datasets.	32	
2	Sep 15 – Sep 19	• Attended project briefing to understand task requirements. • Wrote SQL queries to extract data from multiple tables. • Started data cleaning using Pandas.	40	
3	Sep 22 – Sep 26	• Completed data cleaning and pre-processing. • Performed Exploratory Data Analysis (EDA) to identify initial patterns. • Created basic visualizations (charts, graphs).	40